

Breaking timescales with generative sampling of conformational transitions

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Chenyu Tang^{1,5}, Mayank Prakash Pandey^{1,5}, Cheng Giuseppe Chen¹, Alberto Megías², François Dehez¹ & Christophe Chipot^{1,3,4}

Molecular transitions, including protein folding, allostery and membrane transport, are central to biological functions, yet remain notoriously difficult to simulate. Their intrinsic rarity places them beyond the reach of standard molecular dynamics, whereas enhanced-sampling strategies are computationally demanding and often depend on arbitrarily chosen parameters and variables that bias outcomes^{1–3}. Here we introduce Gen-COMPAS, a generative committor-guided path-sampling framework that reconstructs transition pathways without predefined collective variables and at acceptable computational cost. Gen-COMPAS couples a denoising diffusion probabilistic model, which produces structurally plausible intermediate targets, with committor-based filtering to identify transition states^{4,5}. Short unbiased simulations from these intermediates yield transition-region ensembles at nanosecond-to-submicrosecond aggregate sampling scales for which conventional approaches require orders of magnitude more sampling. Applied to systems ranging from a miniprotein to a pentameric, ligand-gated ion channel, Gen-COMPAS recovers committors, transition states and free-energy landscapes from known end-point structures alone, without predefined reaction coordinates or prior mechanistic knowledge, thereby providing a computationally tractable route to mechanistic insight in biomolecular systems that have so far resisted conventional simulation approaches.

Rare transitions between long-lived metastable states occur in chemical and biomolecular systems and in materials^{6,7}. In molecular systems, they underlie conformational changes, chemical reactions, and recognition and association phenomena central to biological function. Characterizing their transition-state ensembles (TSEs), dominant pathways and free-energy landscapes (FELs) is, therefore, essential for mechanistic understanding, quantitative prediction and rational modulation of molecular function. Yet these events often occur on microsecond-to-second timescales, making their direct observation by brute-force molecular dynamics prohibitively expensive for many systems^{1,8}.

Enhanced-sampling methods² address this limitation by biasing dynamics along reaction coordinates, typically represented by a small set of collective variables (CVs), while reconstructing the thermodynamic observables. Their effectiveness depends on choosing CVs that capture the relevant slow degrees of freedom. Incomplete CVs can conceal metastable states or distort pathway exploration. String-like CVs^{9,10} provide pathway-based descriptions, but their refinement can be costly or they converge to suboptimal pathways when initialized far from the dominant transition tube. Even when informative CVs capture the relevant dynamics, their practical use can pose challenges. In particular, an inadequate bias may drive the system away from thermodynamic equilibrium¹¹ and potentially distort the molecular structure in unintended ways, or it may alter the natural kinetics of barrier crossing.

Transition-path theory^{5,12} identifies the committor q as the optimal reaction coordinate and provides a basis for constructing committor-consistent strings¹³. Deep-learning approaches based on transition-path theory have enabled the study of large conformational transitions^{7,14}, and recent variational formulations^{15–17} allow q and committor-consistent strings to be learned iteratively from biased path-CV sampling. Cartesian-space models¹⁸ further reduce the reliance on an explicit dimensionality reduction. Nevertheless, these approaches require transition data supplied by enhanced sampling.

Generative approaches and approaches guided by molecular dynamics, including Boltzmann generators¹⁹, MDGen²⁰, BioEmu²¹, trajectory predictors²² and transition-path samplers^{23,24}, learn equilibrium ensembles, surrogate dynamics or transition proposals in a data-driven fashion. These advances have motivated methods that combine machine-learned structural proposals with explicit-Hamiltonian dynamics. However, many models rely on extensive pretraining, modified dynamics or pre-existing transition data, although such information is often unavailable for a new rare event. With only a few starting structures, obtaining kinetic and mechanistic quantities, such as committors, transition states and pathways, together with thermodynamic observables, such as reweighted FELs, remains strenuous. This challenge is especially pronounced in heterogeneous biomolecular assemblies involving membrane coupling, ligand binding, domain rearrangements and several slow degrees of freedom.

¹Laboratoire de Physique et Chimie Théoriques (LPCT), Université de Lorraine, CNRS, Nancy, France. ²Complex Systems Group and Department of Applied Mathematics, Universidad Politécnica de Madrid, Madrid, Spain. ³Department of Biochemistry and Molecular Biology, University of Chicago, Chicago, IL, USA. ⁴Theoretical and Computational Biophysics Group, Beckman Institute, and Department of Physics, University of Illinois at Urbana-Champaign, Urbana, IL, USA. ⁵These authors contributed equally: Chenyu Tang, Mayank Prakash Pandey. ✉e-mail: chenyu.tang@univ-lorraine.fr; mayank.pandey@univ-lorraine.fr; chipot@illinois.edu

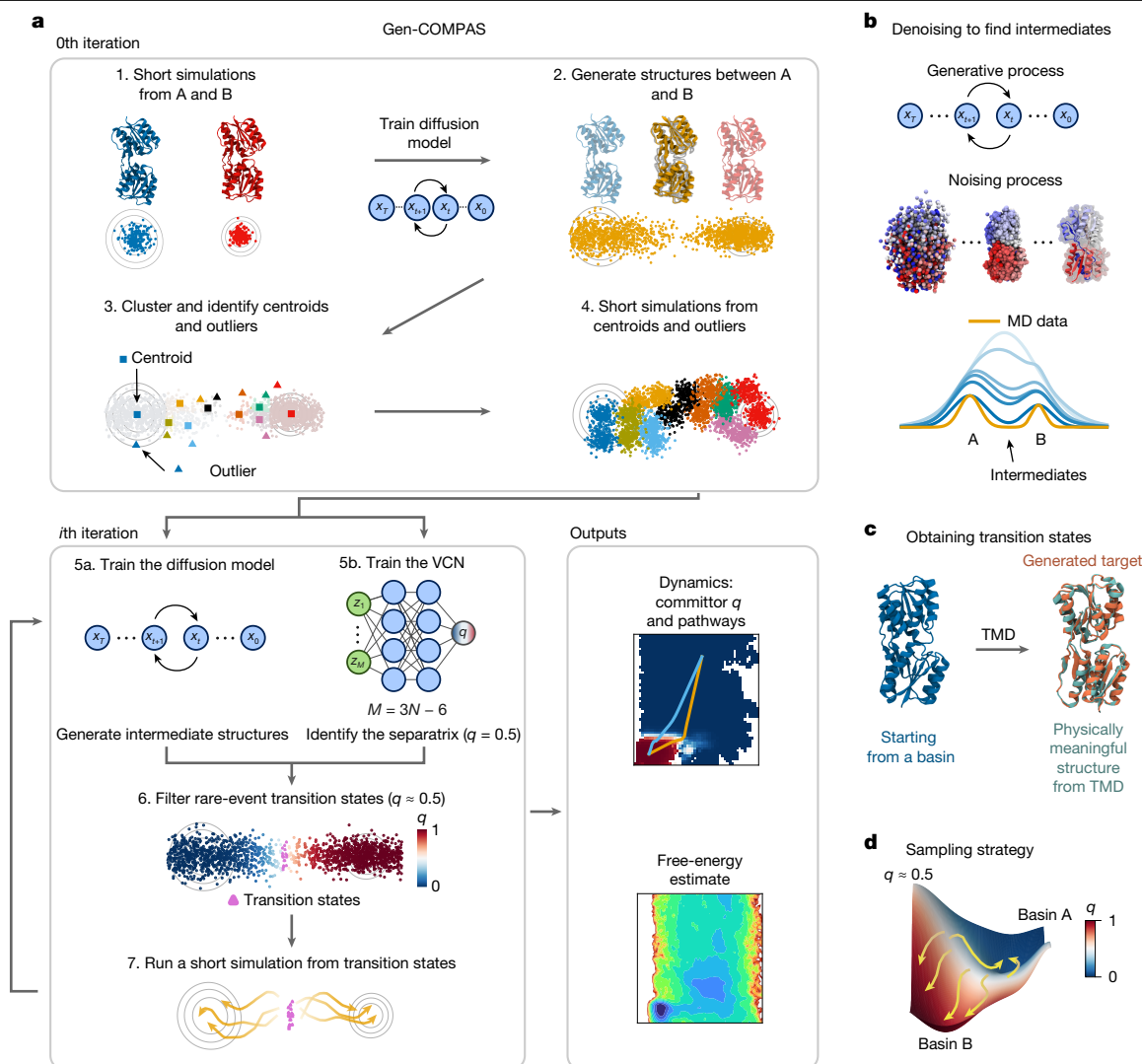


Fig. 1 | Gen-COMPAS in a nutshell. **a**, Overall framework of Gen-COMPAS. Here, x_T denotes the terminal noisy configuration of the diffusion process, x_i an intermediate noisy configuration, and x_0 the generated configuration. N is the number of selected atoms, and $z = (z_1, \dots, z_M)$, with $M = 3N - 6$, denotes the translationally and rotationally reduced configurational representation used

as input to the VCN. **b**, Denoising diffusion model. Training and inference are used to find the intermediates. **c**, Obtaining physically meaningful structures corresponding to transition states using TMD. **d**, Overall sampling strategy between two metastable states of Gen-COMPAS guided by the committor *q* in a schematic FEL.

Building on these advances, we introduce generative committor-guided path sampling (Gen-COMPAS), an end-to-end framework for constructing transition pathways and FELs between known end states under an explicit molecular Hamiltonian. Gen-COMPAS requires endpoint structures and state definitions but no predefined sampling CVs. It generates candidate intermediates, refines them with molecular dynamics and retains configurations through iterative committor learning. The resulting unbiased trajectories are used to estimate TSEs, transition-path ensembles and reweighted FELs in post-selected CVs.

Gen-COMPAS framework

Gen-COMPAS implements an iterative sequence of generative proposal, molecular dynamics refinement and committor-guided selection (Fig. 1a). The workflow begins with short unbiased simulations of a few metastable states. These simulations are typically 1–2 ns long and are initiated from reactant state A and product state B. They provide the initial training data for a diffusion-based generative model⁴. The model proposes intermediate conformations connecting the states (Fig. 1b). In the first iteration, these structures are clustered because a committor

model is not yet available. In subsequent iterations, a high-dimensional committor function, learned directly in conformational space, identifies structures near the separatrix, $q = 1/2$, from which either state is reached first with equal probability (Fig. 1d). Details of the committor learning are provided in Methods and in Supplementary Information.

Because generative proposals may not lie on the physically accessible transition manifold, they are refined under the full molecular Hamiltonian. Two sets of targeted molecular dynamics (TMD) simulations are initiated from states A and B and steered by the structural root-mean-square deviation (RMSD) towards each proposed target (Fig. 1c). Unbiased molecular dynamics trajectories are then launched from the refined configurations, and committor-consistent candidates are retained for subsequent analysis. The resulting data are used to retrain both the diffusion model and the committor predictor. This iterative feedback loop progressively samples the transition region between A and B. TMD trajectories are used only for refinement and are excluded from the kinetic and thermodynamic estimators. After refinement and diagnostic convergence checks, the cumulative unbiased trajectories support the identification of transition states, the projection of committor maps onto post-selected interpretable CVs,

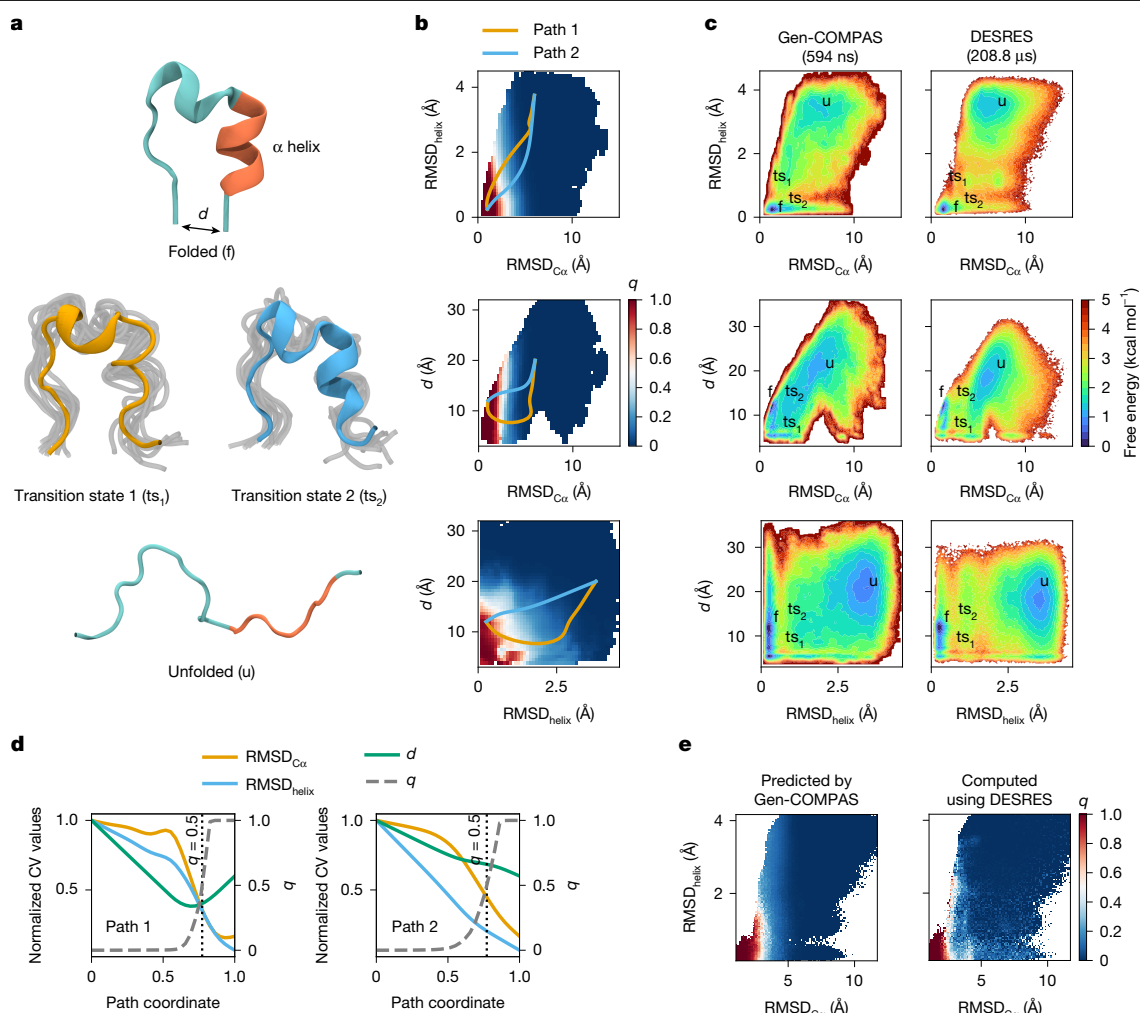


Fig. 2 | Gen-COMPAS applied to a Trp-cage fast-folding protein.

a, Representative structures of Trp-cage in the folded, transition and unfolded states. The α helix and the end-to-end distance d are highlighted. **b**, Learned committor q projected onto the RMSD of distance with respect to all the C_α atoms, the RMSD with respect to the C_α atoms of the α helix and the end-to-end distance. The two sampled pathways are also depicted. **c**, Reweighted FEL

projected onto the same CVs, compared semi-quantitatively with the DESRES simulations⁶. **d**, Normalized values of the CVs and corresponding committor values along the pathways. **e**, Comparison between the committor map of the DESRES trajectory predicted by Gen-COMPAS and the committor computed by counting first-arrival events to unfolded and folded states.

the extraction of committor-consistent strings and the estimation of FELs using RiteWeight²⁵.

Gen-COMPAS is an exploratory sampling workflow rather than a complete replacement for converged equilibrium sampling^{2,3} or transition-path sampling (TPS)^{26,27}. The results of the framework depend on the end-point structures and state definitions, the force field, the transition-region coverage, and the accuracy of the generative and committor models. Generated structures become physically interpretable only after molecular dynamics refinement and validation. Global free-energy differences remain estimates unless convergence is independently established, as for free-energy calculations generally³. The workflow can also miss pathways that are not connected to the supplied basins or are not reached by the proposal and refinement cycle.

No CVs are required to drive generative proposal or committor learning, and interpretable CVs are introduced only for post-simulation projection. Sampling is concentrated near the separatrix, and GPU implementations permit application to large biomolecular systems. In contrast to several protein-focused generative approaches^{20,21}, the joint coordinate representation includes proteins, ligands and other components of heterogeneous biomolecular assemblies. The generated configurations are used as proposals for explicit-Hamiltonian

TMD and unbiased molecular dynamics rather than treated directly as equilibrium samples.

We evaluated Gen-COMPAS on systems of increasing complexity. These applications included *N*-acetyl-*N'*-methylalaninamide and trialanine isomerization in vacuum, Trp-cage folding and the binding-upon-folding of ribose-binding protein in an explicit solvent, the mitochondrial ADP/ATP carrier (AAC) conformational transition, the V_o rotation, and the nicotinic acetylcholine receptor (nAChR) activation in membrane environments. The small benchmark systems reproduced established FELs, committor maps, TSEs and transition pathways^{16–18,28,29}. The ribose-binding protein test checked for coupled ligand–protein interactions, whereas the V_o test looked for transitions involving several metastable states^{30,31}. Results for tests with *N*-acetyl-*N'*-methylalaninamide, trialanine, ribose-binding protein and V_o as well as the sampling procedure and computational requirements for each application are reported in Supplementary Information.

Trp-cage folding

We first benchmarked Gen-COMPAS on the fast-folding Trp-cage mini-protein, whose small size permits atomistic simulation while retaining the coupled secondary- and tertiary-structure rearrangements characteristic of protein folding^{32–34}.

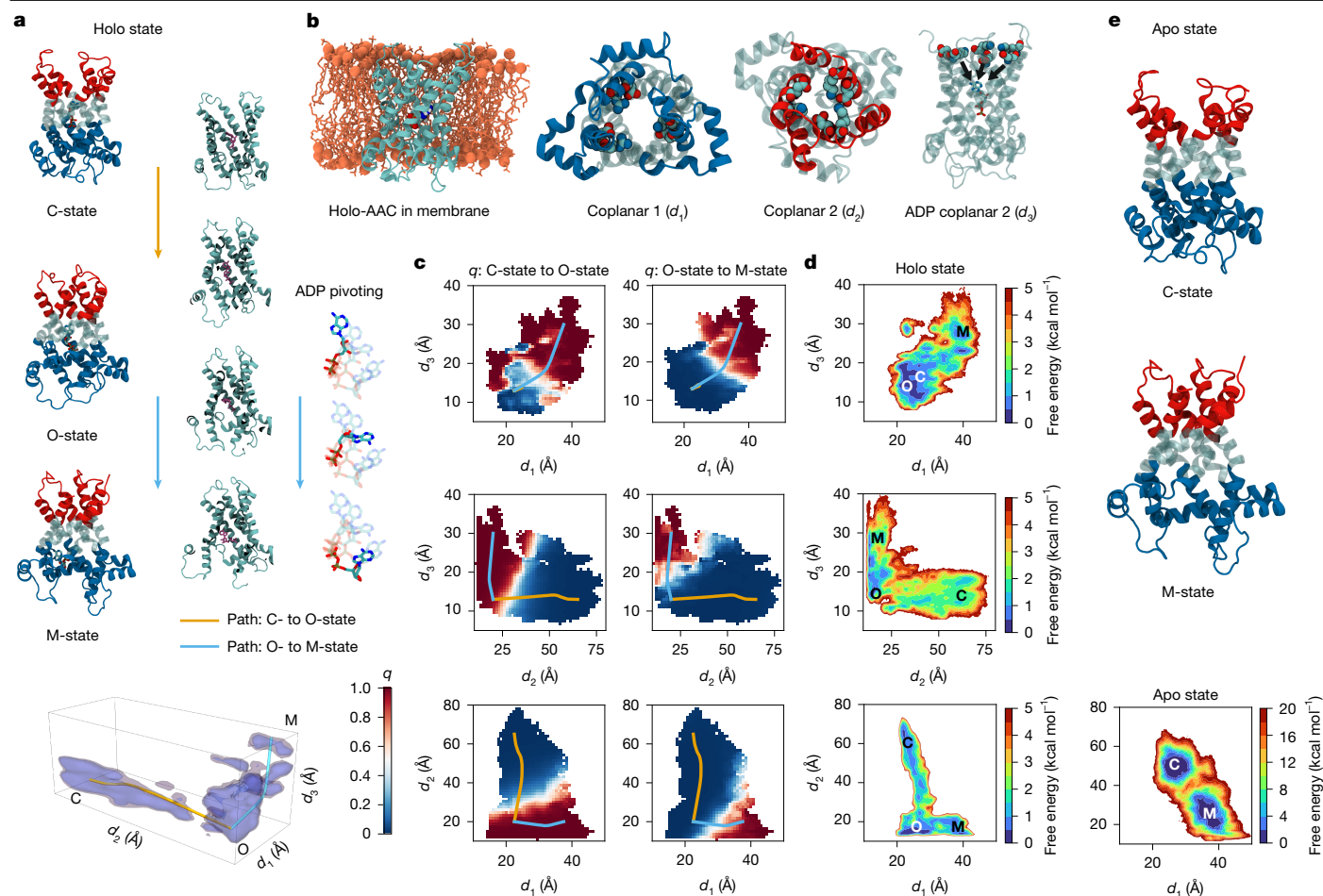


Fig. 3 | Gen-COMPAS applied to the mitochondrial AAC. **a**, The three metastable states, transition states and transition pathway found by Gen-COMPAS from the C-state to the M-state through the O-state of holo AAC (ADP³⁻ bound). The pivoting motion of ADP³⁻ from the O-state to the M-state is also shown. Bottom, the conformational transition pathway C → O → M as well as the basins of the estimated FEL in the three-dimensional CV space. **b**, Holo state

of AAC in the membrane and the CVs describing the AAC conformational transition: d_1 , d_2 and d_3 . **c**, Committors and committor-consistent pathways connecting the three states. **d**, The FEL estimates projected onto two dimensions. **e**, The two metastable states of apo-AAC and its FEL estimate projected onto d_1 and d_2 .

Gen-COMPAS identified candidate transition-state structures, recovered the principal basins of the folding FEL and learned the folding committor (Fig. 2a–c,e). The estimated FEL and predicted committor are broadly consistent with those from microsecond-timescale DESRES simulations⁶. We treat this comparison as semi-quantitative because the force fields, finite sampling and reweighting procedures were not identical. Aimless shooting independently validated the learned committor, with agreement comparable with that reported in earlier committor-based TPS studies¹⁴ (Supplementary Information). Our analysis of the committor and TSE further revealed a bifurcated folding mechanism with two dominant competing pathways (Fig. 2b,d), consistent with previous TPS studies in an explicit solvent²⁷. One route involves early helix nucleation followed by core consolidation. The other initiates with the hydrophobic collapse of key tertiary contacts around the central tryptophan residue, with helix formation lagging behind³².

For Trp-cage, Gen-COMPAS used 594 ns of aggregate unbiased sampling, whereas the conventional molecular dynamics trajectory used for semi-quantitative validation was 208 μ s long. This approximately 350-fold difference in trajectory length reflects a comparison of sampling budgets rather than the asymptotic decay of statistical error or wall-clock time, as a shorter conventional molecular dynamics simulation could yield comparable observables. The learned committor, thereby, estimates the probability of folding before unfolding without requiring a folding event to occur in a single continuous trajectory.

Unravelling the complex machinery of the mitochondrial AAC

The AAC is a mitochondrial transporter that maintains the cellular energy balance by exchanging cytosolic ADP³⁻ for matrix ATP⁴⁻ through a strict one-to-one antiport mechanism^{35,36}. Because the inner mitochondrial membrane is impermeable to nucleotides, AAC provides the essential entry route for the ADP³⁻ required for oxidative phosphorylation. The transporter operates through an alternating-access mechanism that cycles between a cytoplasmic-open (C-) state that binds ADP³⁻ and a matrix-open (M-) state that releases ADP³⁻ and binds ATP for export³⁷ (Fig. 3a). Biochemical and computational studies have proposed a transient occluded (O-) state in which the substrate is enclosed within the central cavity and shielded from both sides of the membrane^{38,39}. However, direct observation of the full transport pathway has remained difficult because conventional molecular dynamics rarely samples transitions between these intermediate configurations⁴⁰.

Using Gen-COMPAS, we reconstructed the ADP³⁻ transport pathway and identified a clear C → O → M mechanism (Fig. 3a,c). In this pathway, ADP³⁻ first binds in the cytoplasmic-open state, becomes transiently occluded through coordinated rearrangements of the transmembrane helices and is subsequently released into the matrix as AAC adopts the matrix-open conformation. These results support the long-hypothesized occluded configuration as a necessary step in the transport cycle.

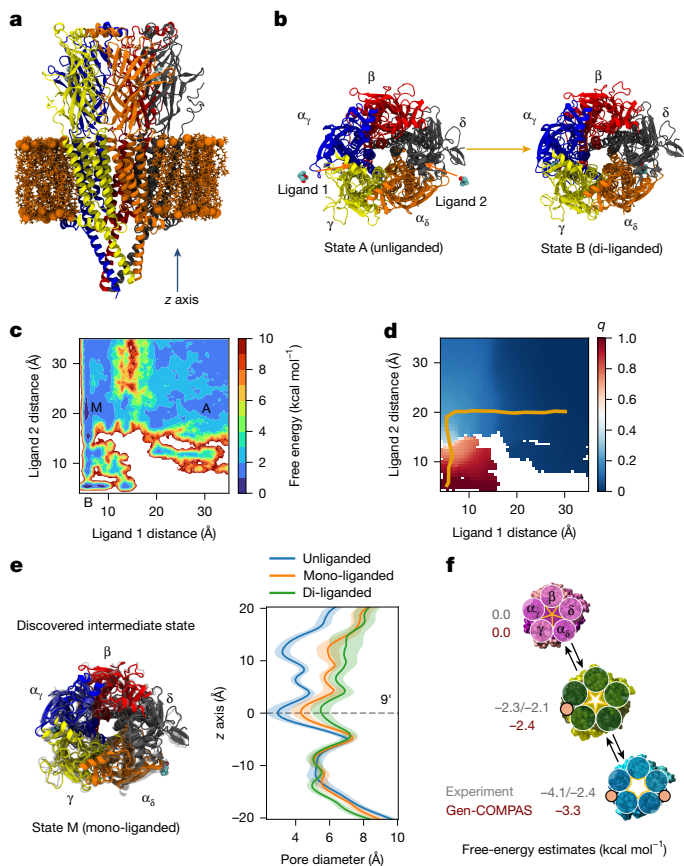


Fig. 4 | Gen-COMPAS applied to the nAChR. **a**, The di-liganded and pore-active nAChR in the membrane. **b**, Unliganded and pore-closed state of nAChR (state A) and di-liganded and pore-active state of nAChR (state B). The activation is triggered by the binding of the two ACh molecules. **c**, **d**, The FEL (**c**) and the committor and committor-consistent pathway (**d**) projected onto two dimensions (the distances between ligands with respect to their binding sites). **e**, Left, the discovered intermediate mono-liganded and pore-pre-active state clustered from Gen-COMPAS. The translucent underlying structure is the experimentally solved mono-liganded state (PDB 9E3G). Right, the pore diameter along the z axis of the simulation box aligned by the 9° position of the channel for unliganded, mono-liganded and di-liganded states. **f**, Proposed gating mechanism. Association of the first ACh molecule at the α/γ interface induces a pre-active configuration of the pore. Subsequent binding of the second ACh molecule at the α/δ interface completes the activation and stabilizes the active channel. The apparent basin free-energy differences were estimated by integrating the corresponding basins on the FELs and are compared semi-quantitatively with experimental kinetic-model-based estimates.

To characterize the thermodynamics of this transition, we projected the sampling onto three interpretable CVs defined post-simulation (Fig. 3b): two describing the opening of the cytoplasmic and matrix salt-bridge networks and a third describing the position of ADP³⁻ within the central cavity. The resulting FEL revealed distinct basins corresponding to the C-, O- and M-states separated by barriers consistent with helix-gating motions of the transporter^{38,41} (Fig. 3d). A committor analysis further identified the occluded region as the transition checkpoint, from which the probability of progressing towards the matrix-open state dominates (Fig. 3c).

To probe the role of the substrate, we also performed Gen-COMPAS simulations for the apo form of AAC (Fig. 3e). In contrast to the holo system, where the C → O and O → M barriers are modest (approximately 2.5 kcal mol⁻¹ and approximately 2 kcal mol⁻¹), the apo landscape lacks the occluded basin and exhibits a much higher barrier (approximately 10 kcal mol⁻¹) between the C- and M-states. This result indicates that

the transition is strongly disfavoured without ADP³⁻, highlighting the role of the substrate in stabilizing the transport-competent pathway^{35,38}.

Beyond elucidating the mechanism that underlies the functionality of ACC, our findings can link transporter dynamics to mitochondrial physiology and disease. As AAC is central to bioenergetics, dysfunction in ADP transport can compromise ATP production and contribute to diseases ranging from mitochondrial myopathies to neurodegeneration. By confirming and quantifying the occluded state, Gen-COMPAS has provided a mechanistic foundation for understanding how mutations and inhibitors can disrupt ADP³⁻ transport, paving the way for the rational modulation of mitochondrial function in health and pathology while complementing the extensive molecular dynamics simulations previously used to sample complete transition pathways in membrane transporters⁴².

Asymmetric nAChR activation via a mono-liganded intermediate

The muscle-type nAChR is a prototypical pentameric ligand-gated ion channel that mediates fast synaptic transmission at the neuromuscular junction (Fig. 4a). It is a main pharmacological target⁴³. Activation is governed by long-range allosteric coupling between the extracellular ligand-binding domain and the transmembrane pore.

The transition from the resting unliganded state to the di-liganded activated state is difficult to simulate because commonly used symmetry constraints or symmetric CVs can average out subunit-specific motions^{44,45}. Structural studies instead indicate that activation involves symmetry-breaking and asynchronous rearrangements⁴³. Because Gen-COMPAS does not require predefined sampling CVs, it can capture such asymmetric motions directly.

We reconstructed the transition between the unliganded pore-closed state, state A (PDB 9E3F), and the di-liganded pore-active state, state B (PDB 9E3E) (Fig. 4b). The same two explicit acetylcholine (ACh) molecules observed in the di-liganded structure by cryogenic electron microscopy were included at both end points and represented jointly with the receptor. Ligand–protein distances were used only after sampling to project the FEL and define interpretable basins. The resulting FEL and committor analysis revealed the transition pathways connecting the end-point states (Fig. 4c,d). The unbiased trajectories revealed a distinct mono-liganded basin and associated intermediate structures (Fig. 4e). In this configuration, one ACh occupies a binding interface, the second site remains unoccupied and the receptor adopts a partially activated conformation.

This sequential unliganded to mono-liganded to di-liganded pathway is consistent with cryogenic electron microscopy and electrophysiological measurements by Thompson et al.⁴³, which resolved the same three states and showed that agonist binding proceeds sequentially. Both the experimental structures and simulations support an asymmetric mono-liganded intermediate, in which one principal subunit is active-like while neighbouring subunits remain inactive but primed for activation.

Our simulations further indicate that ACh association at the α/γ interface first induces a partial rearrangement and a pre-activated pore configuration. Subsequent binding at the α/δ interface completes the activation and stabilizes the active channel (Fig. 4f). Integrating the FEL basins gives apparent free-energy differences for the simulated finite-sized system that fall within the range inferred from experimental kinetic schemes⁴³. Because no correction to a 1 M ligand standard state was applied, we interpret this comparison as semi-quantitative rather than as an estimate of standard binding free energies.

The recovery of this ordered asymmetric pathway indicates that Gen-COMPAS can resolve coupled ligand occupancy and receptor rearrangement without imposed symmetry or predefined reaction coordinates. The all-heavy-atom representation can describe interactions across heterogeneous components, whereas a subsequent molecular dynamics propagation tests the physical plausibility of the proposed configurations.

Discussion

Gen-COMPAS combines system-specific generative proposals, committor-guided selection and explicit-Hamiltonian molecular dynamics to explore rare-event dynamics from end-point structures and state definitions. In contrast to dataset-based generative approaches, which are limited by the statistical patterns of pre-collected training data^{20,21}, Gen-COMPAS trains on trajectories generated for the system at hand and tests proposed intermediates under its molecular Hamiltonian. The same workflow can, in principle, use higher fidelity Hamiltonians such as ab initio molecular dynamics or machine-learning force fields, subject to their computational cost and accuracy.

Gen-COMPAS is best viewed as an exploratory workflow for mechanistic analysis. The reweighted FELs can reveal metastable basins, barriers and transition regions, but their global equilibrium convergence is not established for every complex system (Supplementary Information). We, therefore, used these FELs to support local and pathway-level observations, and the absolute global A-to-B free-energy differences remained as estimates. Identified TSEs can subsequently seed enhanced sampling^{2,3} or TPS^{26,27} when more rigorous convergence is required.

In summary, Gen-COMPAS provides a route to proposing, refining and validating transition-region structures in high-dimensional conformational landscapes. Refined generative models and committor-guided sampling could extend the workflow to transitions among several metastable states beyond the two-state setting considered here. As generative modelling and molecular dynamics continue to converge, a provocative, albeit legitimate question arises: are we witnessing the beginning of the end for exhaustive microsecond-to-millisecond brute-force simulations and traditional enhanced-sampling strategies? Gen-COMPAS does not entirely replace these approaches but, rather, demonstrates that the essence of rare-event dynamics can be revealed without extreme computational cost, potentially reshaping the future trajectory of molecular simulations.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-026-11025-1>.

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Methods

Overall workflow of Gen-COMPAS

The scope of Gen-COMPAS includes activated processes ranging from chemical reactions and metastable-state dynamics to protein and membrane-transporter conformational changes^{46–49}. Existing strategies for accessing such processes include special-purpose molecular dynamics hardware⁵⁰, established enhanced-sampling methods² and differentiable rare-event simulations⁵¹. Path-based approaches include string methods^{9,10}, TPS²⁶ and dedicated path-sampling software⁵². Data-driven approaches learn CVs using autoencoders or graph neural networks^{53–55}, estimate committors from short trajectories^{56,57}, or combine committor learning with adaptive exploration^{15–18}. Generative approaches instead learn equilibrium ensembles, surrogate trajectories or transition proposals^{19–24,58}.

Gen-COMPAS integrates and adapts established components rather than introducing each component de novo. The foundations are denoising diffusion models⁴, including related score-based formulations⁵⁹, variational committor learning^{15,16,18}, committor-consistent path construction^{13,17}, TMD⁶⁰ and trajectory reweighting with RiteWeight²⁵. The methodological contribution here is their organization into a feedback loop in which a system-specific generative model proposes structures, explicit-Hamiltonian molecular dynamics refines them, the committor selects transition-region targets and the accumulated unbiased trajectories are used to retrain both learned models.

The workflow starts from short unbiased trajectories initiated in the end-point basins A and B . In the first iteration, the diffusion model is trained on these data, and the generated structures are clustered because a committor model is not yet available. TMD trajectories launched from both end points towards each selected target identify configurations accessible from both sides. Unbiased shooting from these configurations supplies transition-region data. From the second iteration onwards, a variational committor network (VCN) is trained on the accumulated trajectories; generated structures with $q \approx 1/2$ are selected, refined by TMD and used to initiate further unbiased trajectories. The cycle is repeated until the committor, TMD consistency and reweighted FEL diagnostics described in Supplementary Information cease to change within the reported uncertainty.

The generated coordinates are, therefore, proposal targets, not equilibrium samples or final transition states. TMD is used only for refinement and is excluded from kinetic and thermodynamic estimators. Those estimators use the subsequent unbiased trajectories. The workflow can miss pathways not connected to the supplied basins or not reached by the proposal and refinement cycle, and the global FEL estimates remain limited by trajectory connectivity and sampling coverage.

Denoising diffusion probabilistic model

The generative component is a denoising diffusion probabilistic model operating on centred three-dimensional atomic coordinates⁴. A fixed forward process adds Gaussian noise according to a variance schedule, and a neural network is trained to predict the clean coordinates from a noised configuration and the diffusion time. At inference, iterative denoising maps Gaussian noise to candidate intermediate conformations. The model is retrained after each sampling cycle so that later proposals incorporate transition-region configurations obtained from explicit-Hamiltonian simulations.

The denoising network jointly represents atoms selected for each molecular component, including ligands, using a covalent-plus-dynamic-neighbour graph. Atom-level continuous-filter message passing⁶¹ is combined with residue-level attention⁶² to couple the local geometry to longer-range rearrangements. Protein and ligand coordinates are denoised together rather than generated and docked separately. The stochastic noise scale is reduced across refinement cycles to shift from broader initial exploration towards proposals closer to the accumulated data. Complete equations, architecture

parameters, noise schedules and per-system settings are provided in Supplementary Information.

Committor function and simulation schemes

Discovering the transition-state and path ensembles. The methodology proposed in this work is designed to study conformational transitions in molecular systems in equilibrium at a fixed temperature T . Within these premises, the collection of all conformations accessible to the molecular process defines the state space $\Omega \subseteq \mathbb{R}^{3N}$, in which two disjoint metastable subsets, $A, B \subset \Omega$, are identified as the reactant and product states, respectively. The overarching goal is to characterize the ensemble of reactive trajectories connecting A to B , together with the corresponding transition structures, so as to infer the microscopic mechanisms underlying the process.

The committor function as the descriptor for transitions. The natural theoretical framework for addressing this problem is transition-path theory, the fundamental ingredient of which is the committor probability q . Formally, the committor is a bounded scalar function, $q: \Omega \rightarrow [0, 1]$, defined for an initial conformation $\mathbf{x}_0 \in \Omega$ as

$$q(\mathbf{x}_0) = \mathbb{P}(\tau_B(\mathbf{x}_0) < \tau_A(\mathbf{x}_0)), \quad (1)$$

where $\tau_S(\mathbf{x}_0) = \inf\{t > 0 | \mathbf{x}(t|\mathbf{x}_0) \in S\}$ denotes the first hitting time of the set $S \subset \Omega$, and $\mathbf{x}(t|\mathbf{x}_0)$ is the trajectory at time t initiated from $\mathbf{x}_0$ under equilibrium dynamics. By construction, the committor satisfies as boundary conditions that $q(\mathbf{x}) = 0$ for $\mathbf{x} \in A$ (certain return to the reactant) and $q(\mathbf{x}) = 1$ for $\mathbf{x} \in B$ (certain arrival at the product). Hence, $0 < q(\mathbf{x}) < 1$ for intermediate conformations.

The committor thus provides a probabilistic parameterization of the progress of the transition. Any reactive trajectory must cross all intermediate values of q , monotonically in the forward direction. Owing to this property, q is often regarded as an ideal reaction coordinate. Knowledge of the committor enables the reconstruction of key kinetic information, including the TSE, transition-path ensembles and reactive currents. Such committor-based transition-path analyses have also been used to characterize other activated molecular processes^{63,64}.

The VCN. The VCN¹⁵ consists of a multilayer perceptron with ℓ layers of n nodes and an exponential linear unit⁶⁵ as the activation function. The VCN is based on a variational principle^{5,15} and a relation that connects the scalar flux J_{AB} to q :

$$J_{AB}[q; \tau] = \frac{C[q; \tau]}{\tau}, \quad C[q; \tau] = \frac{1}{2} \langle (q(\tau) - q(0))^2 \rangle. \quad (2)$$

The idea behind the VCN is to find a function that minimizes the scalar flux $J_{AB}[\cdot, \tau]$ or, equivalently, the correlation functional $C[\cdot, \tau]$ for a predefined time lag τ . Hence, the VCN defines a function q_ω as

$$q_\omega(\mathbf{z}) = \begin{cases} 0, & \text{if } \mathbf{z} \in A, \\ F_\omega(\mathbf{z}), & \text{if } \mathbf{z} \in (A \cup B)^c, \\ 1, & \text{if } \mathbf{z} \in B, \end{cases} \quad (3)$$

where $\mathbf{z}$ refers to the set of CVs that defines the reduced CV subspace $\Omega_{\text{CV}} := \{\mathbf{z} \equiv \mathbf{z}(\mathbf{x}) | \mathbf{x} \in \Omega\}$ of the state space Ω , F_ω is a scalar function based on the parameters ω of the neural network, and the superscript c denotes the set complement in configuration space. Therefore, the neural network scalar function was improved according to an optimization scheme that used an Adam optimizer⁶⁶ to minimize the following loss:

$$\mathcal{L}_{\text{VCN}}[q_\omega] = 2C[q_\omega, \tau]. \quad (4)$$

However, equation (3) does not guarantee the boundedness of the output within the interval $[0, 1]$, as illustrated in the original paper¹⁵,

in which regions with values outside this interval are shown in grey. To address this issue, we added a final layer consisting of a single node with a sigmoid activation function, $\sigma(x) : -1/(1 + e^{-x})$, which strictly constrains the output to the $[0, 1]$ range.

Additionally, following the approach detailed in ref. 16, we have included a loss term on the restraints to enforce a continuous mapping at the boundaries between $(A \cup B)^c$ and the basins. The total loss function is, therefore, given by

$$\mathcal{L}[q_\omega] = 2C[q_{\omega,\tau}] + \lambda (|F_\omega(\mathbf{z})|_{\mathbf{z} \in A}^2 + |F_\omega(\mathbf{z}) - 1|_{\mathbf{z} \in B}^2), \quad (5)$$

where $\lambda \in \mathbb{R}^+$ is a constant set a priori to control the boundary loss. Details of the committor validation are provided in the ‘Committor validation analysis’ section of Supplementary Information. Moreover, the influence of the choice of the lag time τ on the learning process is discussed in detail in refs. 15,16,18.

The Z-matrix-based VCN. A central challenge in constructing an accurate committor function is the choice of the reduced representation of the molecular-state space. In many applications, a handful of physically motivated CVs are selected a priori to define the subspace Ω_{CV} . Although this approach is computationally efficient, it carries an intrinsic risk: If the chosen CVs fail to capture all the slow and relevant degrees of freedom, the learned committor may be systematically biased and incapable of faithfully describing the true transition mechanism. This is a well-known limitation in methods based on reaction coordinates, as an insufficient representation can obscure hidden barriers or introduce spurious mechanistic interpretations.

To overcome this shortcoming, we have adopted in this work a representation of the reduced state space Ω_{CV} in terms of internal coordinates (Z-matrix coordinates) of selected atoms. Specifically, we employ the minimal set of $3N - 6$ coordinates that uniquely describe the conformation of an N -atom system modulo global translations and rotations. This choice is motivated by two key considerations, namely completeness of the representation and absence of a prior CV selection.

First, by parameterizing the molecular conformation in terms of the full set of internal coordinates, the reduced space Ω_{CV} is isomorphic to the original conformational space Ω up to rigid-body motions. Consequently, most of the essential degrees of freedom are kept, and the VCN is provided with the complete information required to learn the committor.

Second, unlike traditional approaches that rely on intuitive and physically interpretable variables (for example, distances, dihedral angles or order parameters), the use of these internal coordinates reduces the need for previous assumptions about the relevant coordinates. The network learns which of the selected internal degrees of freedom are informative for the transition, although the result remains dependent on atom selection and sampling coverage.

The use of internal coordinates is further advantageous for describing reactive processes. Distances, angles and dihedrals naturally encode local structural changes, making them physically interpretable features. Moreover, as all the degrees of freedom are retained, the representation is sufficiently flexible to describe complex collective rearrangements and multi-body mechanisms that cannot be reduced to a few intuitive CVs.

In practice, the Z-matrix representation is constructed such that each atom is defined with respect to a small set of reference atoms, resulting in a hierarchical specification of bond lengths, valence angles and torsions. By construction, the dimensionality of this representation is exactly $3N - 6$ —for nonlinear asymmetric molecular systems—thereby ensuring minimal redundancy. The resulting internal coordinate vector, $\mathbf{z} \in \mathbb{R}^{3N-6}$, is then used as the input to the VCN, defining the reduced space Ω_{CV} .

This design reduces, but does not remove, the risk associated with an insufficient selection of CVs. The accuracy of the resulting

transition-state and path ensembles depends on atom selection, the internal-coordinate representation, the expressivity of the VCN, and the quality and coverage of the sampling.

Targeted molecular dynamics. In practice, once intermediate conformations are generated by the diffusion model and subsequently selected by the committor predictor as probable candidates near the separatrix, they can serve as targets for TMD simulations⁶⁰. TMD is an established technique, whereby the system is driven from an initial conformation $\mathbf{x}(0)$ towards a reference target conformation $\mathbf{x}_{\text{ref}}$ by applying a biasing potential. This biasing potential enforces a controlled reduction of the RMSD between the instantaneous structure and the target, according to a predefined schedule.

The biasing potential is given by

$$V_{\text{TMD}}(t) = \frac{k_{\text{TMD}}}{2N} (\text{RMSD}(t) - \text{RMSD}^*(t))^2, \quad (6)$$

where k_{TMD} is the TMD force constant, N is the number of atoms for the RMSD calculation, $\text{RMSD}(t)$ is the instantaneous RMSD of the system with respect to the target and $\text{RMSD}^*(t)$ is a function that decreases linearly from the initial RMSD at $t = 0$ to zero at the final time step $t = t_f$:

$$\text{RMSD}^*(t) = \text{RMSD}(0) \left(1 - \frac{t}{t_f}\right). \quad (7)$$

This design ensures that the system is gradually steered towards the reference structure while minimizing distortions. In the Gen-COMPAS framework, two sets of TMD simulations are conducted: one starting from the reactant state A and the other from the product state B, both aiming at the same generated intermediate. In this way, the procedure systematically identifies a physically plausible region of conformational space that is accessible from both ends of the transition.

Unbiased shooting. Once the system is brought sufficiently close to the generative intermediate, the biasing potential is removed, and unbiased trajectories are spawned from the last conformation produced by the TMD simulations. These unbiased simulations, devoid of biases, are crucial, as they allow the system to evolve according to unbiased dynamics, thereby providing statistically valid samples of reactive trajectories without artefacts from the steering forces. Importantly, because the starting structures are already preconditioned to lie near the separatrix (as guided by the committor predictor), the probability of observing successful reactive events within the available simulation time is markedly increased, compared with brute-force molecular dynamics.

The iterative use of TMD followed by unbiased shooting thus plays a dual role in the Gen-COMPAS framework. First, it provides an efficient mechanism for refining generative intermediates into physically realizable conformations lying on the separatrix. Additionally, it yields a high-quality ensemble of unbiased trajectories that can be directly used to retrain both the diffusion model and the committor predictor in subsequent iterations.

Molecular systems and the Gen-COMPAS set-up

In this section, we describe the set-up of different applications examined in the main text and the initialization of the Gen-COMPAS framework. All the molecular dynamics simulations described below were performed using the NAMD molecular dynamics engine⁶⁷.

Trp-cage folding in an explicit solvent. The fast-folding 20-residue Trp-cage miniprotein was simulated in explicit TIP3P water with the CHARMM36m force field⁶⁸. The system was equilibrated and subsequently sampled in the NPT ensemble at 290 K and 1 atm using a Langevin thermostat with a damping coefficient of 1 ps^{-1} . Initially, two 2-ns

unbiased simulations were performed from both the folded and unfolded states, which were used to initialize the Gen-COMPAS framework. The CVs for FEL and committor projection include (1) $\text{RMSD}_{\text{helix}}$, defined as the RMSD of the C_α atoms of residues Y3-D9, (2) $\text{RMSD}_{\text{C}_\alpha}$, the RMSD of the C_α atoms of all residues, and (3) d , the distance between the C_α atoms of residues D1 and S20 (Fig. 2a).

AAC in the membrane. The mitochondrial AAC was simulated in a POPC lipid bilayer with explicit TIP3P water molecules using the CHARMM36m force field⁶⁸. The carrier exhibited large-scale conformational transitions among the matrix-open (M), occluded (O) and cytoplasm-open (C) states. The AAC C-state was taken from the reported experimentally resolved structure⁶⁹. The M-state was modelled using MODELLER⁷⁰ by projecting the AAC C-state sequence onto the reported M-state structure⁷¹, whereas the O-state was generated by TMD from the M-state by targeting the CV d_1 towards 20 Å and subsequently relaxing the structure with the substrate bound inside. By performing 2-μs-long simulations of the apo-state AAC (ligand-free) and holo-state AAC (ADP-bound) in the O-state (which are not used during the Gen-COMPAS framework), we discovered that the O-state is stable over microseconds for the holo-state AAC whereas it is not stable within the nanosecond timescale for the apo-state AAC.

The CV d_1 is defined as the sum of distances between three terminal side-chain atom pairs: CE(K32)–CD(D231), CZ(R137)–CE(E29) and CZ(R234)–CD(D134). The CV d_2 is the sum of distances between CE(K95)–CD(D195), CE(K198)–CD(D291) and CE(K294)–CD(D92). Finally, d_3 is defined as the distance between the N6 atom of ADP³⁻ and the centre of mass of the side-chain atoms defining d_1 , namely CE(K32), CD(D231), CZ(R137), CE(E29), CZ(R234) and CD(D134) (Fig. 3b).

All the simulations for AAC were equilibrated in the NPT ensemble at 310 K and 1 atm by employing a Langevin thermostat with a damping coefficient of 1 ps⁻¹. Unbiased simulations of 2 ns were initiated from the M-, O- and C-states embedded in the lipid bilayer. Both apo and holo AAC systems were simulated, and the Gen-COMPAS workflow was subsequently performed to compare their conformational dynamics.

In particular, the holo-state AAC conformational transition involves three metastable states (C, O and M). To account for this multi-state landscape within the Gen-COMPAS framework, we trained two separate committor models during the VCN learning stage. The first committor function was trained to distinguish the C- and O-states, enabling the identification of intermediate configurations along the C → O transition. A second committor function was trained between the O- and M-states to characterize the subsequent O → M transition. For each committor, we characterized the committor-consistent pathways (Fig. 3c). This sequential treatment allowed the sampling procedure to resolve transitions across several metastable basins while maintaining the committor-guided exploration strategy. Beyond the present AAC application, this approach provides a general strategy for applying Gen-COMPAS to systems involving more than two known metastable states, as transitions can be decomposed into successive committor-guided segments connecting adjacent basins along the reaction network.

nAChR activation. The muscle-type nAChR was simulated in an explicit membrane-solvent environment employing the CHARMM36m force field⁶⁸ for the protein and compatible CHARMM parameters for ACh. The simulations were performed in the NPT ensemble at 310 K and 1 atm using a Langevin thermostat with a damping coefficient of 1 ps⁻¹ and periodic boundary conditions in all three dimensions.

Initial configurations for the two end-point states of the activation transition were based on recent cryogenic electron microscopy structures of the muscle-type nAChR⁴³. The unliganded, pore-closed structure (PDB 9E3F) served as the receptor-unbound reactant state (state A), with the same two explicit ACh molecules included in the

system but initially placed in the solvent away from the two binding interfaces to keep the molecular composition identical between the two end points. The di-liganded, pore-active structure complexed with two ACh molecules (PDB 9E3E) defined the product state (state B). Thus, the calculation describes a finite-system transition between a receptor-unbound closed end point and a di-liganded active end point with two explicitly represented ACh molecules. No other ACh bath was included. Consequently, ligand binding and release in this analysis refer to the motion of these two explicit ligands relative to the binding interfaces, rather than to a grand-canonical exchange with a bulk ligand reservoir.

Short unbiased molecular dynamics simulations, each of 2 ns, were performed from both the unliganded, closed and di-liganded, active states to generate the initial dataset required to train the diffusion model and initialize the Gen-COMPAS workflow. The resulting configurations were used to train the generative model that proposed intermediate conformations between the two metastable states. Subsequent sampling followed the iterative generative-committor-guided procedure described above, enabling efficient exploration of the transition region connecting the closed and active conformations of the nAChR channel.

The CVs are defined as the distances between the N1 atom of each bound ACh molecule and the C_α (W149) in the α_1 (ligand 1 distance) and α_2 (ligand 2 distance) subunits of nAChR (Fig. 4b). These distances were used for post-simulation projection, visualization, and basin integration only; they were not used to drive the diffusion model, TMD target generation or VCN training. To characterize the functional state of the channel along the transition, we computed the pore-diameter profile using the HOLE program⁷². For representative structures corresponding to the unliganded, mono-liganded and di-liganded states identified from the Gen-COMPAS trajectories, HOLE quantified the pore diameter along the channel axis (the z direction of the simulation box; Fig. 4e). This analysis allowed us to map the structural changes associated with pore activation and distinguish between closed, pre-active and active channel conformations.

To estimate the free-energy differences between the unliganded, mono-liganded and di-liganded states (Fig. 4f), we integrated the probability density over predefined basins on a three-dimensional FEL. The three CVs used to construct this FEL were (1) x , the perimeter of the pore-forming pentagon defined by the C_α atoms of residues α_1 L251, β L257, δ L265, α_2 L251, and γ L260 at the 9' position of the nAChR, (2) y , the ligand 1 distance, and (3) z , the ligand 2 distance. The basins corresponding to the unliganded (A), mono-liganded (M) and di-liganded (B) states were defined by the following ranges in the three CVs (Å): A: $x \in [35, 41]$, $y \in [15, 50]$, $z \in [15, 50]$; M: $x \in [40, 45]$, $y \in [0, 7]$, $z \in [15, 50]$; B: $x \in [42, 50]$, $y \in [0, 7]$, $z \in [0, 7]$ with $F \in [0, 5]$ kcal mol⁻¹. The free energy of each state was obtained by integrating the Boltzmann-weighted probability within the corresponding basin. For visualization purposes, Fig. 4d shows a two-dimensional projection of the FEL onto the ligand 1 and ligand 2 distances, although the basin integration was performed on the full three-dimensional landscape.

Data availability

Input files of the systems analysed in this work are available at GitHub (<https://github.com/Tangcyu/Gen-COMPAS/tree/main/examples>). Source data are provided with this paper.

Code availability

Gen-COMPAS is available as an installable open-source package at GitHub (<https://github.com/Tangcyu/Gen-COMPAS>). The repository provides installation instructions, a unified command-line interface for executing the complete workflow, ready-to-use configuration files and molecular-simulation inputs for representative systems in the examples

directory. A graphical user interface is also provided to guide users through the available options and generate valid configuration files.

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Author contributions Data curation: C.T. Conceptualization: C.T., M.P.P., C.C. Methodology: C.T., M.P.P., C.C. Software: M.P.P., C.T., C.G.C. Formal analysis: C.T., M.P.P., C.G.C., A.M. Validation: C.T., M.P.P., C.G.C., A.M. Visualization: C.T., M.P.P., C.G.C., A.M. Writing – original draft: C.T., M.P.P., C.G.C., A.M., F.D., C.C. Writing – review & editing: C.T., M.P.P., C.G.C., A.M., F.D., C.C. Supervision: C.C.

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Correspondence and requests for materials should be addressed to Chenyu Tang, Mayank Prakash Pandey or Christophe Chipot.

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